MESA SCHOOL OF BUSINESS

Data Foundations · Pre-Read

Session 2: Zomato

Big Mastery Question: Tag every order 'healthy' or 'we lost money on it' — then tell marketing which discount rule to kill.

Before you come to class

Read this once, slowly. This session is all about TEACHING SHEETS TO MAKE DECISIONS — instead of you eyeballing every row, you'll write a rule once and Sheets applies it to thousands of orders instantly. As before, every table below shows real column letters and row numbers so you know exactly where each formula lives.

The company: Zomato delivers food from restaurants to hungry customers across India. Every order creates one row: which restaurant, what cuisine, how much it cost, how much discount was applied. Some orders make Zomato money after discounts and delivery costs; some don't. Your job this session: build a formula that tags every order as healthy or not — instantly, for every row — so marketing knows which discount rules are bleeding money.

0. The Dataset You'll Be Working With

Zomato's orders table has 10 columns in total (order_id, order_date, city, restaurant_id, cuisine, order_value, discount, delivery_fee, payment_mode, is_first_order). Below is a 6-column, 6-row slice — enough to work through every concept this session:

	A	C	E	F	G	J
1	order_id	city	cuisine	order_value	discount	is_first_order
2	Z2001	Bengaluru	North Indian	450	50	TRUE
3	Z2002	Mumbai	Chinese	320	0	FALSE
4	Z2003	Bengaluru	South Indian	180	20	TRUE
5	Z2004	Delhi	Italian	610	100	FALSE
6	Z2005	Mumbai	Chinese	275	0	TRUE
7	Z2006	Bengaluru	North Indian	90	10	FALSE

Column A = order_id, C = city, E = cuisine, F = order_value, G = discount, J = is_first_order (TRUE means this was the customer's first-ever order). Columns B, D, H, I exist in the real sheet but aren't shown here.

1. IF

Real world reference:

Think of a bouncer outside a club with one simple rule: "if you're on the list, you get in — if not, you don't." One question, two possible outcomes. IF works exactly like that bouncer, but for a spreadsheet cell: it checks one condition and gives you one answer if it's true, another if it's false.

In Zomato's world:

You want to tag every order as "Healthy" or "Lost Money." Let's say any order above ₹300 counts as healthy (it comfortably covers costs). IF checks the order_value in column F for every row and instantly labels it — no manual scanning through hundreds of orders.

Syntax:

```
=IF(logical_test, value_if_true, value_if_false)  Example: =IF(F2>300, "Healthy", "Lost Money")
```

See it in the table:

	A	F	K
1	order_id	order_value	Health Flag
2	Z2001	450	Healthy
3	Z2002	320	Healthy
4	Z2003	180	Lost Money
5	Z2004	610	Healthy
6	Z2005	275	Lost Money
7	Z2006	90	Lost Money

Column K is a brand-new column — the flag Zomato didn't have until you built it. Formula in K2: =IF(F2>300,"Healthy","Lost Money"), then autofilled down.

2. Nested IF

Real world reference:

One bouncer with one rule can only sort people into two groups. But what if you need THREE groups — say, "kids," "teens," and "adults"? You'd need a second bouncer standing behind the first one, asking a follow-up question only to the people the first one let through. That's a NESTED IF — an IF formula tucked inside another IF, for when two outcomes aren't enough.

In Zomato's world:

"Healthy or not" is a good start, but marketing actually wants THREE discount tiers: None, Medium, and High — so they can see exactly how aggressive each order's discount was. That needs two questions asked in sequence, which is exactly what a nested IF does.

Syntax:

```
=IF(test1, result1, IF(test2, result2, result3))  Example: =IF(G2=0,"None",IF(G2<50,"Medium","High"))
```

See it in the table:

	A	G	L
1	order_id	discount	Discount Tier
2	Z2001	50	High
3	Z2002	0	None

4	Z2003	20	Medium
5	Z2004	100	High
6	Z2005	0	None
7	Z2006	10	Medium

The formula first asks "is discount 0?" — if not, it asks a SECOND question ("is it under 50?") before deciding. Two questions, three possible answers.

3. IFS

Real world reference:

Nested IFs work, but stack four or five of them and the formula turns into a wall of brackets that's easy to break. IFS is a tidier way to ask a whole LIST of questions one after another, in plain order, without nesting one inside another — like a checklist instead of a chain of bouncers.

In Zomato's world:

The exact same Discount Tier logic from Nested IF, written as a flat checklist instead: "is discount 0? Then None. Otherwise, is it under 50? Then Medium. Otherwise, High." Same answer, easier to read — and much easier to add a fourth tier later without breaking anything.

Syntax:

```
=IFS(test1, result1, test2, result2, test3, result3, ...)  Example:
=IFS(G2=0,"None", G2<50,"Medium", G2>=50,"High")
```

See it in the table:

	A	G	L
1	**order_id**	**discount**	**Discount Tier**
2	Z2001	50	High
3	Z2002	0	None
4	Z2003	20	Medium
5	Z2004	100	High
6	Z2005	0	None
7	Z2006	10	Medium

Identical output to Nested IF above — IFS is a different way to write the same logic, not a different answer.

4. AND

Real world reference:

Imagine a ride at an amusement park with two rules: you must be tall enough, AND you must be wearing shoes. Miss even ONE of those and you don't get on — both conditions have to be true at the same time. That's exactly what AND checks in a formula.

In Zomato's world:

You want to find the truly best orders — ones that are BOTH big (order_value over ₹300) AND cost nothing in discount. An order that's only big, or only discount-free, doesn't count — it has to satisfy both conditions at once.

Syntax:

```
=AND(condition1, condition2, ...)  Example: =AND(F2>300, G2=0)
```

See it in the table:

	A	F	G	M
1	**order_id**	**order_value**	**discount**	**Best Order?**
2	Z2001	450	50	FALSE
3	Z2002	320	0	TRUE
4	Z2003	180	20	FALSE
5	Z2004	610	100	FALSE
6	Z2005	275	0	FALSE
7	Z2006	90	10	FALSE

Only Z2002 is TRUE — it's the only order that's both above ₹300 AND has zero discount. Z2005 has zero discount too, but at ₹275 it fails the first condition, so the whole thing is FALSE.

5. OR

Real world reference:

Now imagine a different rule: you can enter the park if you have a ticket, OR if you're on the guest list. You only need ONE of those to be true — not both. OR is happy the moment any one of its conditions checks out.

In Zomato's world:

Marketing wants to flag every order that's for Indian food — meaning North Indian OR South Indian cuisine. An order only needs to match ONE of those two cuisines to be flagged, not both at once (which would be impossible anyway — a single order has one cuisine).

Syntax:

```
=OR(condition1, condition2, ...)  Example: =OR(E2="North Indian", E2="South Indian")
```

See it in the table:

	A	E	N
1	**order_id**	**cuisine**	**Is Indian Food?**
2	Z2001	North Indian	TRUE
3	Z2002	Chinese	FALSE
4	Z2003	South Indian	TRUE
5	Z2004	Italian	FALSE
6	Z2005	Chinese	FALSE
7	Z2006	North Indian	TRUE

Any order that matches EITHER cuisine gets flagged TRUE — Chinese and Italian orders match neither, so they're FALSE.

6. NOT

Real world reference:

If a light switch is ON, NOT flips it to OFF — and if it's OFF, NOT flips it to ON. That's all NOT does: it takes whatever TRUE or FALSE answer you already have and flips it to the opposite.

In Zomato's world:

Column J already tells you if an order IS a first-time order (TRUE/FALSE). But you actually want to know the opposite — which orders are from REPEAT customers. Instead of writing a whole new condition, NOT just flips the existing is_first_order flag.

Syntax:

```
=NOT(condition)  Example: =NOT(J2)
```

See it in the table:

	A	J	O
1	order_id	is_first_order	Is Repeat Customer?
2	Z2001	TRUE	FALSE
3	Z2002	FALSE	TRUE
4	Z2003	TRUE	FALSE
5	Z2004	FALSE	TRUE
6	Z2005	TRUE	FALSE
7	Z2006	FALSE	TRUE

Every TRUE in column J becomes FALSE in column O, and every FALSE becomes TRUE — NOT is a perfect mirror.

7. IFERROR

Real world reference:

Imagine a vending machine that, instead of jamming and flashing an ugly error code when something goes wrong, politely hands you a note saying "try again" instead. IFERROR is that polite note — it catches an ugly formula error and replaces it with something readable, instead of letting it break your sheet.

In Zomato's world:

You're calculating each order's discount rate as discount ÷ order_value. That works fine — until one row has an order_value of 0 (a cancelled or test order), and dividing by zero throws a scary #DIV/0! error that can break other formulas built on top of it. IFERROR catches that and shows "N/A" instead, so your sheet stays clean and usable.

Syntax:

```
=IFERROR(formula, value_if_error)  Example: =IFERROR(G2/F2, "N/A")
```

See it in the table:

	A	G	F	P
1	order_id	discount	order_value	Discount Rate

2	Z2001	50	450	0.11
3	Z2002	0	320	0
4	Z2003	20	180	0.11
5	Z2007	0	0	N/A

Z2007 is a cancelled order with order_value = 0. Without IFERROR, column P would show #DIV/0! and could break any formula that reads from it. With IFERROR, it shows a clean "N/A" instead.

Coming to class

You now know how to make Sheets ask a question and act on the answer — for one row, or a thousand. In class, we'll build the real Health Flag on the full orders table, then use it to figure out which discount rule marketing should kill.